

AYAN CHANDRASEKARAN

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EDUCATION

University of Michigan SAT: 1570

Ann Arbor, MI

B.S. Computer Science (College of Engineering), Minor in Music

Expected Graduation May 2028

Languages: Python, C/C++, Java, TypeScript/JavaScript, Swift, C# (Unity), SQL, HTML/CSS

Testing & Tooling: Automated testing (Playwright, XCTest), unit/integration test suites, Git, code review, Agile/Scrum, debugging

Frameworks & Platforms: PyTorch, React, Flutter, Node, Postgres, REST APIs, WebSockets, Figma, Claude Code, Cursor, Linux

Relevant Coursework: Data Structures & Algorithms, OOP [EECS 280/281], Machine Learning [EECS 245], Computer Security [EECS 388], Computer Organization/Assembly [EECS 370], Computer Vision [EECS 442], Operating Systems [EECS 482]

EXPERIENCE

Smith+Nephew

Pittsburgh, PA

Intern, Machine Learning and Computer Vision for Surgical Robotics

Summer 2026 - Presented at company-wide AI Conference

May 2026 – August 2026

- Built a full-stack **Python/PyTorch** CNN/VLM classification app for Pressure Ulcer staging, diagnosis, and product recommendation. Wrote HTML/CSS/JS frontend, migrated Flask backend to FastAPI
- Redesigned inference to run models concurrently rather than sequentially, cutting full-diagnosis time from minutes to **3–5s** with the first result instant; verified with **Playwright** tests covering GUI and concurrent API behavior
- Trained/fine-tuned a **ResNet-50** classifier achieving **human clinician-level bedside accuracy (74% vs 67%)** on high-confidence predictions using a single image with **no patient history necessary**
- Built an **experiment harness using custom agent skills** to run architecture, hyperparameter, and augmentation **sweeps in parallel, scoring each run rather than accepting generated output**, cutting dangerous **under-staging errors by 21%**

Summer 2025

June 2025 – August 2025

- Cleaned/analyzed surgical data from **6,500+ cases across 60 surgeons**, holding interviews and cross-functional meetings
- Trained Gradient Boosted Trees and Probabilistic models to predict the Total-Knee surgery time based on surgeon/patient characteristics

Google Student Developers Club @ Michigan, Project Manager

Ann Arbor, MI

Project Manager, building software solutions for charity organizations

October 2025 – Present

- Led wellness-check system for **HomeRise**, a shelter network with **18 locations across San Francisco**, by building an **automated alert pipeline** in Python that notifies staff when a resident hasn't checked in within 72 hours
- Designed the check-in form UI/UX and submission pipeline (HTML/CSS/JS, form validation, backend intake) feeding the alert system, **coordinating with an 8-person team** to ship the feature into production

Emanate, Startup (<https://www.emanate.studio/>)

Ann Arbor, MI

Founder, producing interactive visuals for concerts and events on campus

August 2025 – Present

- Developed C++/Python software enabling musicians to generate audio with **granular synthesis** using **hand gestures**
- Implemented a **projection mapping** system for musical installations on campus, including a **playable projected piano**

PROJECTS

Multiplayer Piano, (<https://bl0opy.github.io/multiplayer-piano/>) [TypeScript, WebSockets, Distributed Systems]

Remote

Real-time collaborative web instrument on a serverless edge runtime

May 2026 – Present

- Built a browser piano **multiple people play together** in real time over **WebSockets**, with one **Cloudflare Durable Object** per room acting as the server for note events, presence, and live cursors
- Wrote an **automated test suite** in **Playwright** driving two real browser contexts into a single room to **verify multi-client sync**; built the client in vanilla **TypeScript** (Web Audio sampler with per-player voice isolation, cursor sync)
- Used the **WebSocket Hibernation API** so idle rooms are **evicted from memory** and cost zero compute

synqd, iOS App (<https://mysynqd.com/>) [Full-Stack, Swift, SQL, Version Control]

Remote

Founder, notification-based long-distance couples app to see current timezone and send custom messages

May 2026 – Present

- Built iOS app on **Supabase** backend, keeping home/lock screen widgets in sync with a partner's **live timezone/location**
- Engineered **real-time sync backend** (Postgres, edge functions, APN) that pushes partner location updates in **under 1 second**
- Launched on TestFlight to **30 users in the first week** and shipped their requests across 4+ builds

SOLO.AI, iOS App [Swift, Python, On-Device ML, Local AI/Agent]

Remote

Founder, privacy-first iOS chat app that scrubs personal data on-device before any cloud LLM sees it

April 2026 – Present

- Architected **hybrid inference pipeline** running a local language model fully on-device via a custom Swift package, **cutting response start time from ~20s to ~2.7s**
- Designed a **three-tier data-protection system** where the on-device model detects Personally Identifiable Information (PII), **reformulates sensitive prompts** to minimum necessary context, and **escalates high-risk** ones to user review
- Implemented **streaming responses** and **local voice input**, validating the privacy system with an **89-test automated suite**

ACTIVITIES & LEADERSHIP

Y Combinator Startup School

San Francisco, CA

Accepted from 30,000+ applicants

August 2026

Journal of Emerging Investigators

Chicago, IL

[Published Research](#) - Application of MEMS Pressure Sensor to locate a person in a skyscraper

June 2021 – August 2021